Linear Regression - VisionGATE Questions

GATE Data Science and Artificial Intelligence (GATE DA)

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Document compiled for GATE DA Preparation

1 Linear Algebra Foundations

Q1. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. Which of the following matrix operations are validly defined? (Select all that apply)

(A) AB

(B) $B^\top A^\top$

(C) $A^\top B$

(D) $ABB^\top$

Q2. Let $x, y \in \mathbb{R}^n$ be column vectors and $A \in \mathbb{R}^{n \times n}$ be a real matrix. Which of the following expressions is *always* identically equal to $x^\top A y$?

(A) $y^\top A^\top x$

(B) $y^\top A x$

(C) $x^\top A^\top y$

(D) Both A and B

Q3. Let $x, a \in \mathbb{R}^n$ be column vectors and $W \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Which of the following gradient identities with respect to x are correct? (Select all that apply)

(A) $\nabla_x (a^\top x) = a$

(B) $\nabla_x (x^\top W x) = Wx$

(C) $\nabla_x (x^\top W x) = 2Wx$

(D) $\nabla_x (x^\top x) = 2x$

Q4. Let $X \in \mathbb{R}^{m \times n}$. Which of the following statements is FALSE?

(A) $\text{rank}(X^\top X) = \text{rank}(X)$

(B) The null space (kernel) of X is identical to the null space of $X^\top X$.

(C) If X has linearly independent columns, then $X^\top X$ is invertible.

(D) $\text{rank}(X^\top X)$ is always equal to m.

Q5. Consider a square matrix $A \in \mathbb{R}^{n \times n}$. Which of the following conditions independently guarantee that A is invertible? (Select all that apply)

(A) The determinant of A is non-zero ($\det(A) \neq 0$).

(B) The columns of A form a basis for $\mathbb{R}^n$.

(C) A has exactly n distinct real eigenvalues.

(D) The null space of A contains only the zero vector.

2 Model Formulation

Q1. In order to concisely represent a linear regression model with a bias term (i.e., $y = Xw + b$) as a single matrix-vector multiplication $y = X_{\text{aug}} w_{\text{aug}}$, what transformations must be applied to the design matrix $X \in \mathbb{R}^{N \times D}$ and the weight vector $w \in \mathbb{R}^D$? (Select all that apply)

(A) A column of ones is appended to the design matrix X.

(B) A row of ones is appended to the design matrix X.

(C) The weight vector w is expanded by one dimension to include the bias scalar b.

(D) The target vector y is augmented with a column of ones.

Q2. In a multiple linear regression model formulated as $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$, what is the precise interpretation of the estimated coefficient β_1?

(A) The expected change in y for a one-unit increase in x_1, ignoring the effect of x_2.

(B) The correlation between the feature x_1 and the target y.

(C) The expected change in y for a one-unit increase in x_1, assuming x_2 is held constant.

(D) The proportion of total variance in y that is exclusively explained by x_1.

Q3. Which of the following statements regarding polynomial regression are true? (Select all that apply)

(A) Polynomial regression is considered a linear model because the prediction is a linear combination of the model parameters (weights).

(B) Polynomial regression is considered a non-linear model because the loss function becomes non-convex with respect to the weights.

(C) By utilizing a polynomial feature mapping, the model can learn non-linear relationships or decision boundaries in the original feature space.

(D) Polynomial regression cannot be solved using the standard Ordinary Least Squares (OLS) closed-form solution.

Q4. Which of the following mathematical formulations can be solved using standard linear regression techniques by defining an appropriate feature transformation $\phi(x)$? (Select all that apply)

(A) $y = w_0 + w_1 x + w_2 x^2 + w_3 x^3$

(B) $y = w_0 + w_1 \sin(x) + w_2 \cos(x)$

(C) $y = w_0 + \exp(w_1 x)$

(D) $y = w_0 + w_1 (x_1 \times x_2)$

Q5. Suppose you are training an unregularized Ordinary Least Squares (OLS) regression model. If you scale a specific feature x_1 in your dataset by multiplying it by a constant $c > 0$ (so $x_1^{\text{new}} = c \cdot x_1$), how will the corresponding newly estimated coefficient $\hat{\beta}_1^{\text{new}}$ change compared to the original $\hat{\beta}_1$?

(A) It will be multiplied by c.

(B) It will be divided by c.

(C) It will remain completely unchanged because OLS is scale-invariant.

(D) It will be divided by c^2.

Q6. Suppose you are fitting a linear regression model but you use a standard, non-augmented design matrix $X \in \mathbb{R}^{N \times D}$ and solve for $w = (X^\top X)^{-1} X^\top y$. What geometric restriction are you forcing upon the resulting prediction hyperplane?

(A) It must be orthogonal to the target vector y.

(B) It must be strictly horizontal (zero slope) across all feature dimensions.

(C) It must pass exactly through the origin $(0, 0, \ldots, 0)$.

(D) It will perfectly interpolate all data points, regardless of N.

Q7. Let $X \in \mathbb{R}^{N \times D}$ be a design matrix without a bias column. You create an augmented design matrix X_{aug} by appending a column of ones. Which of the following statements are mathematically true? (Select all that apply)

(A) X_{aug} has dimensions $N \times (D + 1)$.

(B) The maximum possible rank of X_{aug} is $D + 1$ (assuming $N \geq D + 1$).

(C) If the original matrix X already contained a feature column where every entry was the constant 5, then $X_{\text{aug}}^\top X_{\text{aug}}$ will be non-invertible.

(D) Augmenting the design matrix requires the target vector $y \in \mathbb{R}^N$ to also be augmented with an extra 1.

Q8. You fit a linear regression model to predict a house's price (y). One of the features is a binary variable x_{pool}, where $x_{\text{pool}} = 1$ if the house has a pool, and 0 otherwise. What is the interpretation of the estimated coefficient $\hat{\beta}_{\text{pool}}$?

(A) The average price of all houses with a pool in the dataset.

(B) The expected difference in price between a house with a pool and a house without a pool, assuming all other features in the model are held constant.

(C) The probability that a house has a pool, given its price.

(D) The percentage increase in the house's price due to the presence of a pool.

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Q9. In a multiple linear regression model, you discover that two features, x_1 and x_2, are highly correlated (multicollinearity). How does this impact the interpretation of their respective coefficients, β_1 and β_2? (Select all that apply)

(A) It becomes difficult to isolate the individual effect of x_1 because holding x_2 constant while changing x_1 is not reflective of the real data distribution.

(B) The true values of β_1 and β_2 become mathematically impossible to compute, as the OLS algorithm will fail to converge.

(C) The estimated coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ may become highly unstable and exhibit large standard errors.

(D) The coefficients will automatically shrink to zero to compensate for the correlation.

3 Loss Functions

Q1. In the comparison of the L_1 and L_2 losses, which property do the two losses *not* both satisfy?

(A) Convexity
(B) Symmetry
(C) Differentiability
(D) Non-negativity
(E) Monotonicity

Q2. Why is convexity a desirable property of a loss function?

(A) It guarantees that the loss is differentiable everywhere
(B) It guarantees that any local minimum is also the global minimum
(C) It guarantees that the loss is bounded above
(D) It guarantees that the solution is unique for any dataset

Q3. A fit produces residuals of magnitude $1, 1, 1, 1$ and 10. What share of the total loss does the single large residual account for under L_1 and under L_2 respectively?

(A) About 71% under L_1, about 96% under L_2
(B) About 96% under L_1, about 71% under L_2
(C) 20% under both, since it is one of five points
(D) About 50% under L_1, about 90% under L_2

Q4. Given that L_1 is not differentiable at zero, why is it still used in practice?

(A) It converges faster with gradient descent
(B) It has a simpler closed-form solution
(C) It is more robust to outliers, since it does not amplify large errors
(D) It is the only one of the two that is convex

Q5. What exactly goes wrong with the L_1 loss at zero error?

(A) The derivative is zero, so the update stalls before reaching the minimum
(B) The derivative is undefined — it jumps from -1 to $+1$ with no value at the corner
(C) The derivative becomes infinite, so the step size explodes
(D) The derivative is well defined, but the loss stops being convex there

Q6. Why can we not simply use $\sum_i (y^{(i)} - \hat{y}^{(i)})$ as the loss?

(A) The sum is not differentiable

(B) Positive and negative errors cancel, so a poor fit can score zero

(C) The sum is not convex in $\mathbf{w}$

(D) The sum grows with the number of data points

Q7. What does the monotonicity requirement on a loss function state?

(A) The loss never decreases as training proceeds

(B) Over-prediction and under-prediction are penalised equally

(C) A prediction that is closer to the true value always incurs a smaller loss

(D) The loss increases without bound as the error grows

Q8. What is the effect of writing the objective as $J(\mathbf{w}) = \frac{1}{2n}\sum_i (y^{(i)} - \hat{y}^{(i)})^2$ rather than as the plain sum of squares?

(A) It changes the location of the minimum, so $\mathbf{w}$ must be rescaled afterwards

(B) It makes the loss convex, which the raw sum is not

(C) It leaves the minimiser unchanged but alters the gradient magnitude, and so the useful range of α

(D) It has no effect at all on the optimisation

Q9. A loss that penalises predicting 10 when the truth is 8 exactly as much as predicting 6 when the truth is 8 is exhibiting which property?

(A) Monotonicity

(B) Symmetry

(C) Convexity

(D) Zero at perfection

4 Closed Form Solution

Q1. To derive the normal equations for Ordinary Least Squares, we minimize the residual sum of squares $L(w) = (y - Xw)^\top (y - Xw)$. Which of the following represents the correct gradient $\nabla_w L(w)$ set to zero?

(A) $-Xy^\top + X^\top Xw = 0$

(B) $-2X^\top y + 2X^\top Xw = 0$

(C) $-2Xy + 2XX^\top w = 0$

(D) $2X^\top y - X^\top Xw = 0$

Q2. Consider a dataset with a single feature and no intercept, where the feature matrix is $X = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and the target vector is $y = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. What is the exact closed-form OLS estimate for the weight w?

(A) 11/5

(B) 5/11

(C) 7/5

(D) 2

Q3. Let $r = y - X\hat{w}$ be the residual vector from an OLS regression where $\hat{w}$ is the optimal closed-form solution. Which of the following statements are *always* true regardless of whether the model has an intercept? (Select all that apply)

(A) $X^\top r = 0$

(B) $\hat{y}^\top r = 0$

(C) $\sum_{i=1}^{n} r_i = 0$

(D) $y^\top r = 0$

Q4. Let $H = X(X^\top X)^{-1}X^\top$ be the hat matrix (projection matrix) in OLS regression. Which of the following properties accurately describe H? (Select all that apply)

(A) H is an idempotent matrix (i.e., $H^2 = H$).

(B) H is a symmetric matrix (i.e., $H^\top = H$).

(C) $Hy = r$, where r is the residual vector.

(D) The diagonal elements h_{ii} represent the leverage of the i-th data point.

Q5. Let $X \in \mathbb{R}^{n \times d}$ be a feature matrix with n samples and d features (where $n \gg d$). What is the standard time complexity for computing the closed-form OLS solution $\hat{w} = (X^\top X)^{-1}X^\top y$?

(A) $O(n^2 d + d^3)$

(B) $O(n^3 + d^3)$

(C) $O(nd^2 + d^3)$

(D) $O(nd + d^2)$

Q6. Consider the OLS closed-form solution $\hat{w} = (X^\top X)^{-1} X^\top y$. Which of the following conditions guarantee that a *unique* solution for $\hat{w}$ exists? (Select all that apply)

(A) The feature matrix X has full column rank.

(B) The number of samples n is strictly less than the number of features d.

(C) $X^\top X$ is a positive definite matrix.

(D) The columns of X are linearly independent.

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5 Gradient Descent

Q1. Consider the objective function $f(x) = x^2 - 4x$. You are performing standard Gradient Descent to find its minimum. If your starting point is $x_0 = 0$ and your learning rate is $\eta = 0.1$, what is the value of the parameter after one iteration (x_1)?

(A) 0.4

(B) -0.4

(C) 0.8

(D) 4.0

Q2. Which of the following statements about the learning rate η in gradient descent are correct? (Select all that apply)

(A) If η is set too large, the algorithm may overshoot the minimum and diverge, causing the loss to increase.

(B) If η is set very small, convergence is slow, but it guarantees finding the global minimum for any non-convex function.

(C) A learning rate schedule (e.g., step decay or exponential decay) can help the model converge more precisely by reducing η as it approaches the minimum.

(D) The optimal learning rate is a universal constant that must always fall strictly between 0 and 1.

Q3. Which of the following are valid stopping criteria for Gradient Descent, and correctly state their potential failure modes? (Select all that apply)

(A) Stopping when the gradient norm $\|\nabla L(\theta)\| < \epsilon$ can prematurely halt training if the algorithm hits a saddle point in a non-convex loss landscape.

(B) Stopping when the training loss stops decreasing guarantees the model has optimal generalization to unseen data.

(C) Stopping when the parameter update magnitude $\|\theta_t - \theta_{t-1}\| < \epsilon$ can fail (stop too early) if the learning rate decays too aggressively.

(D) Early stopping based on validation loss can fail if the validation set is too small or unrepresentative of the true data distribution.

Q4. Suppose you are training a Linear Regression model on a dataset of 10,000 samples using Mini-Batch Gradient Descent. You choose a batch size of 100. How many total parameter updates will the model perform after exactly 5 epochs?

(A) 5

(B) 100

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(C) 500

(D) 50,000

Q5. When comparing Batch Gradient Descent (BGD), Stochastic Gradient Descent (SGD), and Mini-Batch Gradient Descent (MBGD), which statements are true? (Select all that apply)

(A) BGD computes the exact gradient of the empirical loss, leading to a stable trajectory, but it is computationally expensive per step on large datasets.

(B) SGD computes the gradient on a single data point, causing high variance in the trajectory which can occasionally help escape shallow local minima.

(C) MBGD generally exhibits higher variance in its gradient estimates than pure SGD.

(D) MBGD leverages modern hardware (like GPUs) better than pure SGD because it can compute gradients for multiple examples simultaneously via vectorized operations.

Q6. In the context of OLS Linear Regression, under what circumstances would you strongly prefer using Gradient Descent over the exact closed-form solution $\hat{w} = (X^\top X)^{-1} X^\top y$? (Select all that apply)

(A) When the number of features d is extremely large (e.g., $d > 1,000,000$), making the inversion of $X^\top X$ computationally prohibitive.

(B) When you are operating in an "online learning" environment where new data arrives as a continuous stream over time.

(C) When you need the absolute exact mathematical minimum without any tolerance for approximation error.

(D) When the feature matrix X is perfectly orthogonal.

Q7. Fit $\hat{y} = wx$ to the points $(1, 3)$ and $(2, 5)$, starting at $w = 1$ with $\alpha = 0.1$. Using $\dfrac{\partial J}{\partial w} = -\dfrac{1}{n} \sum_i \left(y^{(i)} - \hat{y}^{(i)}\right) x^{(i)}$, what is w after one step?

(A) $w = 0.6$

(B) $w = 1.4$

(C) $w = 0.2$

(D) $w = 1.8$

Q8. Run gradient descent on $J = 5w_1^2 + w_2^2$ from $(1, 1)$ with $\alpha = 0.1$. Where does one step land?

(A) $(0,\ 0.8)$

(B) $(0.8,\ 0)$

(C) (0.5, 0.9)
(D) (0.9, 0.98)

Q9. What is a weakness of stopping when the gradient norm falls below a fixed threshold?

(A) The gradient norm is never exactly zero, so it can never trigger
(B) It stops too late, since the gradient vanishes only at the exact optimum
(C) The threshold is not scale-free — rescaling the loss or the features changes what it means
(D) It requires the loss to be non-convex to work

Q10. With $n = 1000$ and a mini-batch size of 50, how many weight updates occur over 10 epochs?

(A) 10
(B) 200
(C) 10,000
(D) 50

Q11. How does the per-epoch cost of stochastic gradient descent compare with that of batch gradient descent?

(A) SGD is n times cheaper, since it uses one example per step
(B) They cost the same, $O(nd)$ — the difference is one exact step versus n rough ones
(C) Batch is cheaper, since it updates only once
(D) SGD is more expensive by a factor of d

Q12. You have $n = 1{,}000{,}000$ rows and $d = 20$ features, all in memory. Which method is preferable?

(A) The closed form, since d is small and the cost is dominated by a manageable nd^2 term
(B) Gradient descent, since n is too large for the closed form
(C) Gradient descent, since the d^3 term is prohibitive
(D) Neither — the problem is too large to solve

Q13. Which of the following statements about stopping criteria are correct? *(Choose all the correct answers.)*

(A) A fixed iteration budget is simple but may stop short of convergence or waste steps after it
(B) Small change in loss can fire on a flat plateau far from the optimum

(C) A fixed gradient-norm threshold is not scale-free

(D) Relative thresholds are more portable across problems than absolute ones

(E) Monitoring validation loss instead can stop training before overfitting sets in

(F) A small gradient norm guarantees you are near the global minimum for any loss function

Q14. With n examples and mini-batch size B, which of the following are correct? *(Choose all the correct answers.)*

(A) An epoch is one full pass through the training data

(B) An epoch contains $\lceil n/B \rceil$ updates

(C) Batch gradient descent performs exactly one update per epoch

(D) Pure SGD performs n updates per epoch

(E) All three variants perform the same number of gradient *evaluations* per epoch

(F) "Iteration" and "epoch" are interchangeable terms

Q15. Which of the following statements about batch, stochastic and mini-batch gradient descent are correct? *(Choose all the correct answers.)*

(A) All three cost $O(nd)$ of gradient work per epoch

(B) Batch gradient descent gives the exact gradient, so its loss curve decreases monotonically with a suitable α

(C) SGD's per-example gradients are unbiased estimates of the full gradient

(D) Mini-batching reduces gradient noise by averaging, and suits vectorised hardware

(E) SGD needs the whole dataset in memory at once

(F) A decaying learning rate lets SGD settle rather than bounce indefinitely

Q16. You have data pairs $(y_i, x_i)_{i=1:n}$ where $x \in \mathbb{R}^{14}$ and you perform least squares linear regression to learn a function of the form $y = w_0 + x^\top w$. What is the minimum number of samples required for this to be possible?

Q17. You want to approximate your data as a quadratic function using least squares with a polynomial of the form $y = w_0 + \sum_{r=1}^{p} w_r x^r$. What value should you set p to?

Q18. You have data pairs $(y_i, x_i)_{i=1:17}$ where $x \in \mathbb{R}$ and you perform least squares polynomial regression to learn a function of the form $y = w_0 + \sum_{r=1}^{p} w_r x^r$. What is the maximum value you can set p to?

Q19. Which of the following will likely give a sparse solution for w? *(Select all that apply)*

(A) $\arg\min_w \|y - Xw\|_2^2 + \lambda\|w\|_{1/2}$

(B) $\arg\min_w \|y - Xw\|_1 + \lambda\|w\|_2^2$

(C) $\arg\min_w \|y - Xw\|_2^2 + \lambda\|w\|_3^3$

(D) $\arg\min_w \|y - Xw\|_1 + \lambda\|w\|_{3/4}$

Q20. For an optimization problem of the form $\arg\min_w \|y - Xw\|^2 + \lambda\|w\|_p$, the values of p for which we can NOT guarantee an optimal solution are:

(A) $p > 2$

(B) $p < 1$

(C) $1 < p < 2$

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6 Feature Scaling

Q1. A feature ranges from 10 to 50 in the training set. What is the min-max scaled value of $x = 30$?

(A) 0.6
(B) 0.5
(C) 0.4
(D) 0.3

Q2. A training feature has mean 70 and standard deviation 10. What is the z-score of $x = 85$?

(A) 0.15
(B) 8.5
(C) 1.5
(D) 15

Q3. What is the key difference between min-max scaling and standardization?

(A) Min-max makes the distribution normal; standardization does not
(B) Min-max output is bounded to $[0, 1]$; standardized output is unbounded
(C) Standardization is a non-linear transformation; min-max is linear
(D) Standardization removes outliers; min-max preserves them

Q4. A feature has values mostly between 1 and 10, plus one recording error of 1000. What happens under min-max scaling?

(A) Nothing — min-max scaling is robust to outliers
(B) The outlier maps to 0 and everything else spreads across $[0, 1]$
(C) The other values are all squashed into a narrow band near 0, losing resolution
(D) The transformation fails, since the range is undefined

Q5. One feature has values near 1 and another near 1000. What is the rough condition number of $X^\top X$?

(A) About 1000, matching the ratio of the feature scales
(B) About 10^6, since the entries of $X^\top X$ involve products of feature values
(C) About 1, since the condition number is unaffected by scaling
(D) About 2000, the sum of the two scales

Q6. Why does feature scaling speed up gradient descent?

(A) It moves the minimum to a better location

(B) It makes the loss convex, which it otherwise is not

(C) It lowers the condition number, so one α suits all coordinates and the gradient points closer to the minimum

(D) It reduces the per-iteration cost from $O(nd)$ to $O(n)$

Q7. A feature with standard deviation σ is standardized. Under plain OLS, what happens to its coefficient?

(A) It is multiplied by σ

(B) It is divided by σ

(C) It is unchanged

(D) It is multiplied by σ^2

Q8. After standardizing all features, what does the fitted intercept equal?

(A) It becomes zero

(B) It becomes the mean of y

(C) It is unchanged

(D) It becomes the sum of the feature means

Q9. Plain OLS is invariant to feature rescaling, yet ridge regression is not. Why?

(A) Because $X^\top X + \lambda I$ is otherwise singular

(B) Because ridge has no closed-form solution on unscaled data

(C) Because the penalty acts on the raw coefficient magnitudes, so features in large units get shrunk arbitrarily harder

(D) Because ridge requires all features to lie in $[0, 1]$

Q10. You compute μ and σ over the combined training and test data before splitting. What is wrong with this?

(A) Nothing — scaling is a deterministic transformation, so it cannot leak

(B) Test-set information enters training, so the reported test error is optimistically biased

(C) The training features will no longer have exactly zero mean

(D) The coefficients become uninterpretable

Q11. You fit a min-max scaler on the training set. A test observation exceeds the training maximum. What happens?

(A) It is clipped to 1.0

(B) The scaler is refitted to include it

(C) It maps to a value above 1, which correctly signals extrapolation beyond the training range

(D) It causes the transformation to fail